

CONSTRUCTION OS

PRD_HSE_Module

HEALTH, SAFETY & ENVIRONMENT

Full Product Requirements Document

Enterprise operational safety, environmental control, incident governance and field compliance

Field	Value
Version	1.0
Status	Implementation-ready specification
Date	2026-08-05
Audience	Product, UX/UI, Frontend, Backend, QA, DevOps, Security, Data and Platform Architecture
Platform	Construction OS / Business Control Platform
Access strategy	Full capability first; detailed role restrictions applied through platform authorization without redesign

Permanent product rule

HSE governs whether people, equipment and construction activities are safe and environmentally compliant before, during and after work. Work Progress records completion; QA/QC verifies quality; Tasks execute corrective work; Assets govern equipment; Documents preserve evidence.

1. Document Control

Field	Value
Document	PRD_HSE_Module - Health, Safety & Environment
Primary pages	HSE Dashboard, Site HSE Page, HSE Plan, Hazards, Risk Assessments, Permits, Inspections, Observations, Incidents, Investigations, Corrective Actions, Stop Work, Inductions, Toolbox Talks, Emergency, Environment, Contractor HSE
Connected modules	Group, Company, Projects, Work Progress, QA/QC, Assets, Inventory, HR, Teams, Tasks, Documents, Contracts, Procurement, Calendar, Finance, Notifications, Approvals
Core source-of-truth rule	HSE owns operational safety and environmental control records. Connected modules retain ownership of people, assets, work completion, quality acceptance, files, legal terms and accounting.
Platform Configuration	Mandatory at module, page, section, widget, feature, field, workflow, report and action level.
Audit rule	Issued, submitted, activated and finalized HSE records are immutable. Corrections use revision, addendum, reopening or superseding records.
External integrations	Not required for the initial self-contained release. Architecture remains integration-ready.

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3. Executive Summary

The HSE Module is the live operational control system for health, safety and environmental governance across every company, project, site, contractor, worker, asset and construction activity managed by Construction OS. It must prevent unsafe work, provide immediate field reporting, preserve evidence, enforce safety gates and create a complete audit trail from hazard identification through verified corrective action.

The module is not a repository of inspection forms. It is an enterprise domain with its own lifecycles, validation services, mobile workflows, reporting model and cross-module safety controls.

Business outcome

A company must be able to determine, at any moment, whether work is safe to begin, what risk controls are active, which permits are valid, which workers and assets are eligible, what incidents occurred, and whether corrective actions are truly effective.

3.1 Core outcomes

- Prevent uncontrolled high-risk work from starting.
- Provide fast mobile reporting for hazards, incidents, near misses and observations.
- Establish a single operational source of truth for permits, risk assessments, inspections and HSE actions.
- Connect safety readiness directly to Work Progress, Assets, HR, Contractors and Calendar.
- Preserve defensible evidence and immutable HSE history.
- Measure leading and lagging indicators without turning analytics into automatic employee judgment.
- Support one site or a multi-company, multi-project group architecture.

4. Product Vision and Outcomes

Construction OS must make safety an active operational gate, not a retrospective reporting exercise. The HSE Module must be usable by executives, HSE professionals, site management, supervisors, contractors and workers while preserving strict confidentiality for sensitive incident and health information.

4.1 Product principles

1. Prevention before reporting.
2. Stop-work authority must be fast, visible and auditable.
3. Source modules retain ownership of their business records.
4. Critical safety decisions cannot depend on color alone.
5. Field workflows must function in weak-connectivity conditions.
6. Submitted and issued records remain historically reproducible.
7. Every KPI must drill down to source records.
8. The module must be configurable across construction and other operational industries without changing the core data model.

5. Scope and Non-Goals

5.1 In scope

- HSE dashboards and project/site safety workspaces.
- HSE plans and controlled plan revisions.
- Hazard register and risk assessment engine.
- Method-statement safety context and readiness links.
- Permit-to-work lifecycle and validation gates.
- Safety inspections, observations and corrective actions.
- Incident, near-miss, dangerous-occurrence and environmental-event management.
- Investigation, root cause and evidence integrity.
- Stop-work orders and safety holds.
- Inductions, toolbox talks, competence and worker eligibility.
- PPE requirements and compliance.

- Emergency plans, drills, muster and active emergency mode.
- Environmental aspects, monitoring, waste and spill management.
- Chemical register and hazardous-substance controls.
- Contractor HSE prequalification and performance.
- Mobile, offline, audit, APIs, events, testing and platform configuration.

5.2 Non-goals

- Replacing HR employee records, employment contracts or detailed medical records.
- Replacing Assets maintenance, certification or equipment lifecycle management.
- Replacing QA/QC acceptance of completed work.
- Replacing Work Progress measurement of physical completion.
- Replacing Documents file storage and revision control.
- Replacing Tasks as the execution engine for assigned corrective work.
- Defining jurisdiction-specific legal advice or statutory filing integrations in the initial release.
- Implementing final cross-platform Approvals and Workflow orchestration; the module defines hooks and decision points only.

6. Domain Ownership and Boundaries

Domain / Record	System of Record	HSE responsibility
Hazards, safety risks, permits, observations, incidents, emergency and environmental events	HSE	Full lifecycle, status, evidence, audit and reporting.
Employee identity, employment, leave and sensitive medical records	HR / User Profile	Consume competence, eligibility and operational fitness status only.
Equipment identity, maintenance, inspection certificates and operational state	Assets	Consume safety readiness; issue holds and incident links through events.
Task execution	Tasks	Own required corrective action and verification; synchronize linked Task status.
Physical completion and production	Work Progress	Return work-start safety readiness and active holds.
Quality acceptance and NCRs	QA/QC	Share location, evidence and actions without merging domain ownership.
Documents and controlled files	Documents	Store references, revisions, read requirements and evidence links.
Contractual obligations	Contracts	Track operational compliance and notify about potential breaches.
Stock quantities and consumables	Inventory	Define safety requirements for chemicals and PPE; Inventory owns balances.
Financial postings	Finance	Provide cost references and incident impact context only.
Formal approval routing	Approvals - future	Expose approval points, eligible roles and evidence requirements.

6.1 Hierarchy of controls

9. Eliminate the hazard.
10. Substitute the hazard.
11. Apply engineering controls.
12. Apply administrative controls.
13. Use PPE as the final layer.

Critical rule

The system must not default to PPE when stronger controls are available. Risk-assessment templates must explicitly capture the control hierarchy.

7. Actors and Personas

Actor	Primary objectives	Critical access considerations
Group HSE Director	Group oversight, policy, multi-company analytics, severe-incident governance.	Broad reporting; sensitive investigation access by explicit authorization.
Company HSE Manager	Company plans, standards, audits, contractor controls, escalation.	Company scope with project drill-down.
Project HSE Manager	Operational site readiness, permits, incidents, inspections and actions.	Project/site scope; issue and suspend safety records.
Site HSE Officer	Field inspections, observations, inductions, permits, incident response.	Mobile-first; limited confidential investigation access.
Site / Project Manager	Know whether work can proceed and resolve safety constraints.	Read operational readiness; controlled hold release authority.
Permit Issuer / Receiver	Prepare, validate, activate, suspend and close permits.	Permit-type and site-specific competence required.
Inspector / Investigator	Capture objective evidence and findings.	Record-specific access and evidence integrity.
Supervisor / Foreman	Verify workers, equipment, controls and toolbox talks.	Team/work-package scope.
Worker	View required controls, report hazards and stop unsafe work.	Simple mobile UX; no unnecessary confidential data.
Contractor HSE Representative	Manage contractor compliance, workers, permits and actions.	Own-company records plus explicitly shared site context.
Executive / Auditor	Read dashboards, trends, severe events and audit history.	Read-only, source-drill-down, no operational edits.

8. Information Architecture

8.1 Main navigation

- HSE Dashboard
- Projects and Sites
- Safety Plans
- Hazards and Risks
- Risk Assessments
- Method Statements
- Permits to Work
- Inspections
- Safety Observations
- Incidents and Near Misses
- Corrective Actions
- Stop-Work Orders
- Inductions
- Toolbox Talks
- Training and Competence
- PPE
- Emergency Management
- Environmental Management
- Contractor HSE
- Occupational Exposure
- Reports
- HSE Settings

8.2 Embedded views

- Project Page - HSE summary and active constraints.
- Work Package - safety readiness and linked permits.
- Asset Page - incidents, holds and safety requirements.
- Employee Profile - inductions, competence and operational eligibility.
- Team Page - HSE calendar, toolbox talks and training gaps.
- Contractor / Supplier Page - HSE profile and scorecard.
- Contract Page - HSE obligations and compliance evidence.
- Task Page - linked corrective action.
- Calendar - HSE dates and reminders.
- Documents - controlled HSE files and evidence.

9. HSE Dashboard

9.1 KPI framework

KPI group	Examples	Required behavior
Immediate attention	Critical incidents, active emergency, critical holds, expired high-risk permits, unsafe assets.	Always visible; drill-down to source records; not dismissible without resolution or acknowledgement.
Today	Active permits, planned high-risk work, due inspections, toolbox talks, pending inductions.	Time-zone aware and site-filtered.
Leading indicators	Hazards, observations, inspections, training, toolbox participation, action closure.	Trend and source drill-down.
Lagging indicators	Injuries, LTI, property damage, environmental events, regulatory breaches.	Calculation definition displayed with report.
Readiness	Worker eligibility, asset safety, emergency readiness, HSE plan status.	Calculated by explicit rules and latest authoritative data.

ID	Requirement	Implementation / Rationale
HSE-DASH-001	Every KPI card must open a filtered list of source records.	Prevents unverifiable summary metrics.
HSE-DASH-002	Critical alerts must not rely solely on color.	Accessibility and safety clarity.
HSE-DASH-003	Dashboard state must support company, project, site, contractor and date filters.	Multi-entity operational use.
HSE-DASH-004	Data freshness and last-calculated timestamp must be displayed for computed readiness metrics.	Trust and observability.
HSE-DASH-005	Historical snapshots must remain reproducible after source records change.	Audit and reporting consistency.

10. Site HSE Page

Every active project site requires a dedicated operational workspace that combines current readiness, active controls, workforce eligibility, emergency status and unresolved HSE risk.

10.1 Header

- Project and site identity
- Owning company and principal contractor
- HSE manager and site manager
- Current HSE status
- Current workforce
- Active high-risk work
- Active permits
- Emergency status
- Last inspection
- Days without lost-time incident

10.2 Sections

14. Overview
15. Safety Plan
16. Site Zones
17. Workforce Readiness
18. Active Work
19. Permits
20. Inspections
21. Hazards
22. Incidents
23. Corrective Actions
24. Emergency Readiness
25. Environmental Controls
26. Contractors
27. Documents
28. Calendar
29. Activity Timeline

Site HSE lifecycle

Mobilization	Active	Restricted	Partially Suspended	Suspended	Demobilizing	Closed	Archived
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ID	Requirement	Implementation / Rationale
HSE-SITE-001	Site status changes must be versioned and preserve reason, actor, effective date and affected scope.	No silent operational status changes.
HSE-SITE-002	A site may contain independently controlled buildings, floors, zones and work areas.	Location-based safety control.
HSE-SITE-003	Current workforce must be calculated from authoritative attendance/access data where available and clearly marked as projected when not live.	Avoid false muster certainty.
HSE-SITE-004	Site closures must preserve all historical HSE records and references.	Audit retention.

11. HSE Plan

11.1 Structured plan contents

- Scope and project description
- HSE objectives and applicable requirements
- Responsibilities and organization chart
- Site rules
- Risk-management process
- Permit rules
- Emergency arrangements
- Incident reporting
- Inspection and audit program
- Training and PPE requirements
- Environmental controls
- Waste management
- Contractor requirements
- Reporting schedule
- Communication and document register

HSE Plan lifecycle

Draft	Under Review	Issued	Active	Revised	Superseded	Closed
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ID	Requirement	Implementation / Rationale
HSE-PLAN-001	Each issued plan must reference a controlled document revision and structured plan metadata.	Files remain in Documents; HSE owns plan lifecycle.
HSE-PLAN-002	A revised plan must not overwrite or replace the auditability of prior issued versions.	Historical reproducibility.
HSE-PLAN-003	Plan applicability may be project-wide or location-specific.	Supports phases and complex sites.
HSE-PLAN-004	Required-reading audiences and acknowledgements must use Documents capabilities.	Avoid duplicate read-receipt systems.

12. Hazard Register

12.1 Hazard scope

- Group
- Company
- Project
- Site
- Building
- Floor
- Zone
- Unit
- Work Package
- Activity
- Asset
- Material
- Job Role
- Contractor

12.2 Core fields

Field group	Required fields
Identity	Hazard code, title, description, category, source, status.
Exposure	Location, people exposed, activity, contractor, asset, material.
Consequence	Potential harm, severity, escalation threshold.
Controls	Existing controls, control owner, control effectiveness.
Links	Risk assessments, incidents, observations, work packages, documents, actions.
History	Created by, revisions, status history, last review and archive reason.

- Work at height
- Lifting operations
- Excavation
- Confined space
- Electrical
- Fire and explosion
- Moving machinery
- Vehicle movement
- Structural instability
- Demolition
- Falling objects
- Manual handling
- Noise and vibration
- Dust and chemicals
- Public safety
- Environmental and custom categories

ID	Requirement	Implementation / Rationale
HSE-HAZ-001	Hazard codes must remain unique within the configured business scope.	Stable references in incidents and assessments.
HSE-HAZ-002	Archiving a hazard must not remove it from historical assessments or reports.	Referential integrity.
HSE-HAZ-003	Hazards may be reused by template but site-specific exposure and controls remain separate records.	Avoid accidental global changes.
HSE-HAZ-004	A critical uncontrolled hazard may create or recommend a safety hold according to configuration.	Operational prevention.

13. Risk Assessments

13.1 Supported types

- General Risk Assessment
- Activity Risk Assessment
- Site-Specific Risk Assessment
- JSA / JHA
- Dynamic Risk Assessment
- Chemical / COSHH

- Environmental
- Ergonomic
- Fire
- Lifting
- Custom

13.2 Assessment model

Phase	Data
Context	Activity, scope, location, assessment team, workers exposed, linked work and contractor.
Initial risk	Hazard, likelihood, severity, risk score, rationale.
Controls	Control measure, hierarchy level, owner, due date, evidence, effectiveness.
Residual risk	Residual likelihood, severity, score, acceptance rule.
Operational requirements	Competence, PPE, permits, inspections, emergency and environmental controls.
Review	Validity, review date, trigger conditions, revision and acknowledgements.

Risk Assessment lifecycle

Draft	Assessed	Released	Active	Review Due	Revised	Superseded	Archived
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13.3 Automatic review triggers

- Incident or near miss
- Method or design change
- New equipment or material
- Site-condition change
- Regulatory change
- Control failure
- Periodic review date
- Worker feedback

ID	Requirement	Implementation / Rationale
HSE-RA-001	Risk matrices must be configurable without changing stored historical scores.	Historical reports must retain the matrix used at assessment time.
HSE-RA-002	Residual risk acceptance must record the accepting role and rationale where thresholds require it.	Decision traceability.
HSE-RA-003	A released assessment becomes immutable; revisions create a new version.	Controlled lifecycle.
HSE-RA-004	Dynamic assessments may be short-form but must record context, actor, time and immediate controls.	Field usability with audit.
HSE-RA-005	Permit readiness must validate the exact applicable assessment revision.	Avoid stale safety controls.

14. Method Statements

Documents owns the controlled method-statement file. HSE owns structured safety context, applicability, readiness and relationships to risk assessments, permits and work activities.

- Work sequence
- Responsible contractor and supervisors
- Workforce requirements
- Equipment and materials
- Access and exclusion zones
- Temporary works
- Hazards and controls
- Required permits
- Qualifications and PPE
- Emergency response
- Environmental controls
- Inspection points
- Drawings and revision validity

ID	Requirement	Implementation / Rationale
HSE-MS-001	A new method-statement document revision must trigger HSE readiness re-evaluation.	Safety status cannot be inherited automatically.
HSE-MS-002	Applicability must identify exact projects, locations, activities and effective dates.	Prevents broad accidental use.
HSE-MS-003	Expired or superseded methods cannot satisfy permit validation.	Safety gate integrity.

15. Permit to Work

15.1 Permit types

- Hot Work
- Confined Space
- Excavation
- Work at Height
- Electrical Work
- Isolation / LOTO
- Lifting Operation
- Crane Operation
- Demolition
- Roof Work
- Pressure Testing
- Radiography
- Road Closure
- Live Services
- Chemical Work
- Environmental Discharge
- Night Work
- Custom

15.2 Permit fields

Group	Fields
Identity	Permit number, type, project/site, exact location, activity and work package.

Group	Fields
Parties	Requester, issuer, receiver, competent supervisor, contractor and workers.
Validity	Start/end, shift, extension and revalidation history.
Controls	Risk assessment, method statement, isolations, PPE, equipment, tests, weather and emergency arrangements.
Evidence	Checklists, photos, documents, signatures and closure confirmation.
History	Activation, suspension, revocation, changes, closure and audit events.

Permit lifecycle

Draft	Prepared	Ready for Issue	Issued	Active	Suspended	Revalidated	Closed	Archived
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15.3 Activation gates

- Valid risk assessment and method statement
- Competent and inducted workers
- Current training and certificates
- Valid asset inspections
- Confirmed isolation
- Acceptable weather and area availability
- Emergency arrangements ready
- No conflicting permit
- Required documents present

ID	Requirement	Implementation / Rationale
HSE-PTW-001	Permit activation must be a server-authoritative transaction.	No client-only or offline activation.
HSE-PTW-002	No permit may be activated with an end time earlier than start time or after its validity window.	Temporal validity.
HSE-PTW-003	Conflicting permits must be detected by configurable permit-type, location and time rules.	Prevent hazardous overlap.
HSE-PTW-004	Expiry must automatically remove Active status and emit a domain event.	No stale active permits.
HSE-PTW-005	A stop-work order must suspend affected permits immediately.	Safety hold enforcement.
HSE-PTW-006	Closure must confirm area safe, isolations restored or transferred, and outstanding hazards recorded.	Controlled handback.
HSE-PTW-007	Backdated activation is prohibited.	Audit integrity.

16. Safety Inspections

16.1 Inspection types

- Daily Site
- Weekly HSE
- Management Walk
- Work at Height
- Excavation
- Scaffolding
- Lifting
- Electrical
- Fire
- Housekeeping
- Welfare
- Environmental
- PPE
- Contractor
- Emergency Equipment
- Regulatory
- Custom

16.2 Template capabilities

- Sections and conditional questions
- Pass/fail and measurements
- Risk rating
- Mandatory photo/comment
- Critical questions
- Signatures
- Auto-created observation or corrective action
- Auto-created safety hold
- Asset lockout hook

Inspection outcome

Passed	Passed with Observations	Conditional	Failed	Unsafe	Immediate Action Required	Reinspection Required
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ID	Requirement	Implementation / Rationale
HSE-INSP-001	Inspection templates must be versioned; completed records retain the exact template version used.	Reproducibility.
HSE-INSP-002	Critical question failure must execute configured safety consequences atomically with submission.	No gap between finding and control.
HSE-INSP-003	Photos and measurements may be mandatory by answer condition.	Evidence completeness.
HSE-INSP-004	Reinspection must link to the original failed inspection and unresolved actions.	Traceable closure.

17. Safety Observations

Observations support unsafe conditions, unsafe acts, positive observations and improvement opportunities. The mobile flow must permit rapid capture without weakening the later triage and verification process.

Observation lifecycle

Reported	Triaged	Action Required	Under Correction	Verified	Closed
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17.1 Quick mobile capture

- Scan location QR
- Select category
- Take photo/video
- Voice or text note
- Mark immediate danger
- Identify activity/team
- Anonymous submission where configured

ID	Requirement	Implementation / Rationale
HSE-OBS-001	Immediate-danger observations must offer one-action stop-work escalation.	Fast field response.
HSE-OBS-002	Anonymous reporting must protect reporter identity while preserving abuse controls.	Trust and governance.
HSE-OBS-003	Positive observations must be reportable without being mixed into violation statistics.	Balanced safety culture.
HSE-OBS-004	Closing an observation may require evidence and verifier identity by configuration.	Verified resolution.

18. Incident and Near-Miss Management

18.1 Incident types

- Injury
- Illness
- Fatality
- Near Miss
- First Aid
- Medical Treatment
- Lost-Time Incident
- Restricted Work Case
- Property Damage
- Equipment Accident
- Vehicle Accident
- Fire
- Explosion
- Spill
- Environmental Release
- Public Incident
- Security Incident
- Dangerous Occurrence
- Occupational Exposure

18.2 Initial report

Field group	Minimum data
When and where	Incident date/time, report date/time, site and exact location.
What happened	Type, short description, immediate condition and severity estimate.
People	Affected persons, witnesses and contractor/employer.
Response	Emergency services, first aid, isolation, work stopped and immediate controls.
Context	Activity, work package, asset/equipment and material.
Evidence	Photos, video, attachments and reporter statement.

Incident lifecycle

Reported	Emergency Response	Triage	Under Investigation	Corrective Actions	Verification	Closed
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18.3 Severity

- Minor
- Moderate
- Serious
- Major
- Critical
- Fatal - configurable thresholds and escalation rules

ID	Requirement	Implementation / Rationale
HSE-INC-001	Initial reporting must be possible with the	Field speed without data loss.

ID	Requirement	Implementation / Rationale
	minimum safe dataset and allow controlled completion later.	
HSE-INC-002	Severity changes must preserve prior values, actor and reason.	Audit defensibility.
HSE-INC-003	Severe incidents must trigger configured emergency and executive events.	Immediate escalation.
HSE-INC-004	Incident closure must validate investigation, actions, reporting obligations and evidence requirements.	No premature closure.
HSE-INC-005	Personal and medical fields must support field-level confidentiality.	Privacy.
HSE-INC-006	Deleting incidents is prohibited; invalid records use controlled cancellation with reason.	Immutable history.

19. Investigation and Root Cause

19.1 Investigation workspace

- Investigation team and roles
- Timeline reconstruction
- Witness statements
- Evidence register
- Immediate and underlying causes
- Barrier failures
- Root-cause method
- Findings
- Recommendations
- Regulatory/legal flags
- Action generation
- Final report

19.2 Supported methods

- Five Whys
- Fishbone
- Bow-Tie
- Fault Tree
- Barrier Analysis
- Event and Causal Factor Analysis
- Configurable structured method

19.3 Evidence integrity

Evidence attribute	Requirement
Origin	Uploader, source device, original filename and source record.
Time	Capture time where available, upload time and timezone.
Integrity	File hash and immutable original.
Location	Location metadata only when permitted and available.

Evidence attribute	Requirement
Access	Sensitive-read audit readiness.
Correction	Addendum or replacement link; original remains preserved.

ID	Requirement	Implementation / Rationale
HSE-INV-001	Investigation membership must not automatically expose all medical or legal sections.	Field-level confidentiality.
HSE-INV-002	Witness statements must preserve the original version and controlled addenda.	Evidence integrity.
HSE-INV-003	Final findings become immutable and are superseded only through a formal addendum.	Defensible history.
HSE-INV-004	Root causes must link to corrective actions and effectiveness reviews.	Closed-loop prevention.

20. Corrective and Preventive Actions

HSE owns the required action, due date, verification and effectiveness. Tasks owns execution after conversion. Task completion alone must not close the HSE action.

Corrective Action lifecycle

Open	Assigned	In Progress	Evidence Submitted	Under Verification	Effective	Closed
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- Alternative states: Ineffective, Reopened, Cancelled

Field	Requirement
Source	Incident, inspection, observation, permit, risk assessment, audit, environmental event or contractor review.
Ownership	Responsible person, company, due date, priority and contributors.
Execution	Linked Task, progress projection and evidence.
Verification	Verifier, verification date, result and comments.
Effectiveness	Review period, effectiveness result, reopening and recurrence check.

ID	Requirement	Implementation / Rationale
HSE-CAPA-001	Conversion to Task must be idempotent.	No duplicate execution records.
HSE-CAPA-002	A completed Task must update the action projection but not bypass HSE verification.	Domain ownership.
HSE-CAPA-003	Overdue and critical actions must emit	Operational control.

ID	Requirement	Implementation / Rationale
	escalation-ready events.	
HSE-CAPA-004	Ineffective actions must support reopening or creation of additional actions.	Continuous improvement.

21. Stop Work and Safety Holds

21.1 Hold scope

- Entire project
- Site
- Building
- Floor
- Zone
- Work Package
- Contractor
- Asset
- Permit
- Activity
- Material

Safety Hold lifecycle

Active	Controls in Progress	Ready for Verification	Released
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- Alternative states: Escalated, Extended, Superseded

Field group	Fields
Initiation	Code, reason, hazard, severity, initiator and start time.
Scope	Affected company/project/site/location/work/asset/contractor.
Control	Immediate controls, people notified and blocked actions.
Release	Required conditions, evidence, verifier, release authority, reason and time.
Impact	Duration and links to schedule/cost impact records.

ID	Requirement	Implementation / Rationale
HSE-HOLD-001	Authorized users must be able to issue a hold from mobile in minimal steps.	Stop-work speed.
HSE-HOLD-002	Affected operational records must show the active hold immediately.	Cross-module visibility.
HSE-HOLD-003	Release authority may differ from initiation authority and must be configurable.	Governance.
HSE-HOLD-004	A hold cannot be deleted or silently removed.	Audit safety.
HSE-HOLD-005	Release must validate required evidence	Controlled restart.

ID	Requirement	Implementation / Rationale
	and unresolved critical actions.	

22. Site Inductions

22.1 Induction types

- Company
- Project
- Site
- Visitor
- Contractor
- Role-specific
- High-risk work
- Refresher
- Emergency

Worker eligibility status

Eligible	Eligible with Restrictions	Induction Required	Training Missing	Certificate Expired	Medically Restricted	Suspended	Not Eligible
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ID	Requirement	Implementation / Rationale
HSE-IND-001	Induction records must capture content revision, language, trainer, date, expiry, assessment and acknowledgement.	Proof of exact content delivered.
HSE-IND-002	Site eligibility must be calculated from authoritative induction, competence and operational fitness data.	Reliable access decisions.
HSE-IND-003	Sensitive medical detail must never be returned by eligibility endpoints.	Privacy.
HSE-IND-004	Expired inductions must update eligibility and emit events.	Automatic control.

23. Toolbox Talks

- Topic and linked hazards/incidents
- Site, activity and work package
- Presenter, date/time and language
- Participants, absentees and signatures
- Discussion notes and questions
- Photos and follow-up actions

ID	Requirement	Implementation / Rationale
HSE-TBT-001	Attendance capture must support QR, roster selection and offline mode.	Field efficiency.
HSE-TBT-002	The exact talk content revision must be preserved.	Evidence.
HSE-TBT-003	Follow-up actions may convert to Tasks while remaining linked to the talk.	Execution traceability.
HSE-TBT-004	Participants must not be auto-added to unrelated linked Tasks.	Participant-based task access.

24. Training and Competence

HR and User Profile own qualifications and training records. HSE defines operational competence requirements and consumes verified status to determine work eligibility.

Requirement type	Examples
Qualification	Licensed electrician, certified rigger, scaffolding inspector.
Course	Working at height, confined space, first aid, fire marshal.
Certificate	Operator licence, medical fitness status, external certificate.
Experience	Minimum years, project/site authorization, role assessment.
Restriction	Work type, asset class, location, supervision required.

ID	Requirement	Implementation / Rationale
HSE-COMP-001	Competence rules must be configurable by permit type, activity, asset type and role.	Reusable safety gates.
HSE-COMP-002	Missing competence may block permit activation, site entry, operator assignment or inspection sign-off.	Operational enforcement.
HSE-COMP-003	The eligibility service must return reasons without exposing restricted personal data.	Useful and private.

25. PPE Compliance

HSE owns PPE requirements and compliance. Inventory owns stock quantities. Assets owns individually tracked equipment. HR/User Profile owns the worker relationship.

- Helmet
- Safety footwear
- High visibility
- Gloves
- Eye and hearing protection
- Respiratory protection
- Harness/fall arrest
- Protective clothing
- Gas detector
- Custom PPE

PPE record	Fields
Requirement	Activity, hazard, site, role, standard and mandatory/conditional rule.
Assignment	Worker, item/asset reference, issue date, expiry, condition and size.
Inspection	Interval, last inspection, next inspection and result.
Return / replacement	Return date, replacement reason, disposal and new assignment.

ID	Requirement	Implementation / Rationale
HSE-PPE-001	PPE compliance must be calculated from requirements, assignment, condition, expiry and inspection status.	Accurate readiness.
HSE-PPE-002	Individually tracked PPE must reference Assets rather than duplicate serial records.	Source-of-truth separation.
HSE-PPE-003	Consumable PPE issuance may reference Inventory movements.	Stock traceability.

26. Emergency Management

26.1 Emergency scenarios

- Fire
- Medical emergency
- Structural collapse
- Rescue from height
- Confined-space rescue
- Chemical spill
- Gas release
- Flood
- Severe weather
- Earthquake
- Security threat
- Environmental emergency
- Custom

26.2 Emergency plan data

- Coordinator and response team
- External contacts
- Muster points and escape routes
- Site map
- Emergency equipment
- Hospitals and public services
- Shutdown procedures
- Communication method
- Recovery plan

26.3 Active emergency mode

- Emergency type and location
- Start time and commander
- People affected
- Projected headcount
- Muster status
- Live actions and timeline
- Contact shortcuts
- Resolution status

ID	Requirement	Implementation / Rationale
HSE-EMR-001	Emergency activation must be idempotent and concurrency-safe.	Avoid duplicate active emergencies.
HSE-EMR-002	Projected headcount must be labelled as projected unless based on verified live data.	Prevent false confidence.

ID	Requirement	Implementation / Rationale
HSE-EMR-003	Critical contacts and plans must be available offline to authorized users.	Resilience.
HSE-EMR-004	Emergency resolution must preserve the complete response timeline and actions.	Post-event investigation.

27. Environmental Management

27.1 Aspects and impacts

- Waste
- Water and energy use
- Fuel
- Air emissions and dust
- Noise and vibration
- Soil/water contamination
- Chemicals and spill risk
- Biodiversity
- Traffic
- Light pollution
- Community impact

Record	Core fields
Environmental aspect	Aspect, impact, location, control, responsible person and significance.
Monitoring	Parameter, method, limit, measurement, frequency, device and evidence.
Waste	Stream, classification, quantity, container, carrier, facility, transfer note, recycling and cost reference.
Environmental event	Type, location, containment, impact, remediation, monitoring and closure.

Environmental Event lifecycle

Reported	Contained	Assessed	Remediation	Monitoring	Closed
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ID	Requirement	Implementation / Rationale
HSE-ENV-001	Limit exceedance must emit a critical event and may create a hold or corrective action.	Immediate environmental control.
HSE-ENV-002	Measurement units and limits must be versioned with the record.	Historical interpretation.
HSE-ENV-003	Waste records may link to Inventory items but must not change stock balances directly.	Domain separation.
HSE-ENV-004	Closure must validate remediation and required post-event monitoring.	Prevent premature closure.

28. Chemicals and Hazardous Substances

Field group	Data
Identity	Product, manufacturer, supplier, substance and inventory product link.

Field group	Data
Hazard	Classification, SDS revision, exposure limits and incompatible materials.
Control	Storage, maximum quantity, PPE, ventilation, spill response and disposal.
Location	Site, warehouse, zone and current operational location.
Status	Approved, restricted, expired, prohibited, review due.

ID	Requirement	Implementation / Rationale
HSE-CHEM-001	Safety Data Sheets must be controlled Documents records with revision history.	No duplicate file repository.
HSE-CHEM-002	Using a restricted or expired chemical must trigger configured work-readiness failure.	Operational control.
HSE-CHEM-003	Chemical quantity is read from Inventory where tracked; HSE stores safety thresholds and location context.	Correct source of truth.

29. Contractor HSE

29.1 Contractor profile

- HSE contact
- Policies and plans
- Insurance and certifications
- Worker competence
- Induction compliance
- Incidents and inspections
- Corrective actions
- Permit performance
- Stop-work history
- Environmental performance
- Restrictions and suspension status

29.2 Prequalification

- HSE policy
- Statistics
- Certifications
- Insurance
- Previous incidents
- Training system
- Equipment compliance
- Emergency arrangements
- Subcontractor controls
- References
- Evaluation result

29.3 Scorecard

- Incident rate
- Near-miss reporting
- Inspection score

- Action closure
- Permit compliance
- Training/PPE compliance
- Environmental compliance
- Repeat violations
- Positive observations

ID	Requirement	Implementation / Rationale
HSE-CON-001	Contractor HSE ratings must show calculation components and source records.	Transparent scoring.
HSE-CON-002	Suspension or restriction must preserve effective scope, reason and dates.	Controlled operations.
HSE-CON-003	Procurement owns supplier selection; HSE provides evaluation and readiness data.	Domain boundary.
HSE-CON-004	Historical contractor records remain after project or supplier deactivation.	Audit retention.

30. Occupational Exposure and Fitness Status

HSE may track operational exposure and fitness status required for safe assignment. Detailed diagnoses, treatment and medical records remain restricted to HR/occupational-health systems.

- Exposure type and measurement
- Worker or role group
- Date and duration
- Limit and result
- Protective controls
- Fitness status projection
- Restrictions
- Review date
- Confidential document links

ID	Requirement	Implementation / Rationale
HSE-OH-001	Operational APIs may return Eligible, Restricted, Review Required or Not Eligible, but not medical diagnosis.	Privacy by design.
HSE-OH-002	Exposure records must support confidential access and sensitive-read audit.	High-sensitivity data.
HSE-OH-003	Work restrictions must be effective-dated and scoped.	Accurate assignment control.

31. Cross-Module Integrations

31.1 Assets

- Read operational state, maintenance, certificates, inspection status, location and operator requirements.
- Send incident link, unsafe condition, safety hold, reinspection and operator restriction events.
- Never directly overwrite asset lifecycle or maintenance records.

31.2 Work Progress

- Receive planned activity, location, contractor, workforce, assets, materials, shift and high-risk classification.

- Return risk assessment, method statement, permit, competence, asset readiness and hold status.
- Work Progress remains owner of physical completion.

31.3 QA/QC

- Share location, evidence, contractors, work packages and actions.
- Maintain separate quality and safety results even in a combined inspection experience.

31.4 HR / User Profile

- Consume identity, employer, training, certification and operational fitness projections.
- Write induction, toolbox and site-eligibility links without duplicating employee data.

31.5 Contracts, Procurement and Finance

- Read contractual HSE requirements and contractor/supplier context.
- Provide breach warnings, evidence and operational restrictions.
- Reference costs and claims without creating accounting entries.

31.6 Integration safety gate

Work-start gate

A work package may be blocked when its risk assessment or method statement is missing/expired, required permit is inactive, worker competence is invalid, critical asset compliance is missing, a hazard is uncontrolled, or a safety hold is active.

32. Documents and Evidence Governance

HSE must use the central Documents Module for all files. HSE stores typed relationships, applicability, issue state and evidence metadata; it does not create a second file repository.

32.1 Recommended project folder structure

- 01 HSE Plan
- 02 Policies and Procedures
- 03 Risk Assessments
- 04 Method Statements
- 05 Permits
- 06 Inspections
- 07 Incidents
- 08 Investigations
- 09 Toolbox Talks
- 10 Inductions
- 11 Training
- 12 PPE
- 13 Emergency
- 14 Environmental
- 15 Contractor HSE
- 16 Reports

32.2 Required document capabilities

- Document Passport
- Revision control
- Issue state
- Required reading and receipts
- Expiry
- Watermarks
- Dependency warnings
- References/shortcuts without duplication
- Audit history

ID	Requirement	Implementation / Rationale
HSE-DOC-001	Every document link must identify role, applicability and required revision.	Correct safety context.
HSE-DOC-002	Deleting or archiving a linked document must trigger dependency warnings.	Prevent broken safety records.
HSE-DOC-003	Incident evidence originals must remain immutable.	Evidence integrity.
HSE-DOC-004	Final reports may be generated as controlled PDFs and stored in Documents.	Governed output.

33. Calendar and Reminder Integration

- Permit start/end
- Risk-assessment review
- Inspection
- Toolbox talk
- Induction
- Training/certificate expiry
- Emergency drill
- Environmental monitoring
- Corrective-action due date
- Contractor review

Ownership rule

Calendar displays synchronized projections and deep links. HSE remains the source of truth and all rescheduling or status changes occur in HSE.

34. Notification and Workflow Readiness

34.1 Notification-ready events

- Critical Incident Reported
- Emergency Activated
- Stop Work Issued
- Permit Expiring or Suspended
- High-Risk Work Starting
- Inspection Failed
- Corrective Action Overdue
- Training/Induction Expiring
- Asset Safety Non-Compliant
- Environmental Limit Exceeded
- Emergency Drill Due

34.2 Approval hooks

- Decision point
- Eligible roles
- Required evidence
- Threshold
- Due date
- Escalation hook
- Approval reference
- Decision outcome

The final Approvals & Workflow Engine is intentionally deferred. HSE must not hard-code a separate workflow engine, but all decision points must be integration-ready.

35. Global Search, Filters and Saved Views

35.1 Searchable records

- Sites, plans, hazards, assessments, permits, inspections, observations, incidents, investigations, actions, holds, inductions, toolbox talks, emergencies, environmental events, chemicals and contractors

35.2 Common filters

- Company, project, site and location
- Status and severity
- Category and type
- Contractor and responsible person
- Date range and expiry
- Overdue / due soon
- Confidentiality
- Linked asset/work package
- Personal starred records

ID	Requirement	Implementation / Rationale
HSE-SRCH-001	Search results must respect record, section and field permissions.	No confidentiality leakage.
HSE-SRCH-002	Saved views are private by default and	User control.

ID	Requirement	Implementation / Rationale
	may be shared only through explicit action.	
HSE-SRCH-003	Archived records remain searchable when archive filters are enabled.	Historical access.
HSE-SRCH-004	Critical records must be indexed with near-real-time updates.	Operational freshness.

36. Reporting and Analytics

36.1 Leading indicators

- Hazards and near misses reported
- Inspections completed
- Observations
- Toolbox participation
- Training compliance
- Permit compliance
- Corrective-action closure
- Management walks
- Positive observations
- Emergency drills

36.2 Lagging indicators

- Injuries
- Lost-time incidents
- Medical-treatment cases
- Restricted-work cases
- Fatalities
- Property damage
- Environmental events
- Regulatory breaches
- Work stoppage duration

36.3 Reports

- Site HSE
- Contractor HSE
- Permit
- Inspection
- Incident
- Investigation
- Corrective Action
- Training Matrix
- PPE Compliance
- Environmental
- Emergency Readiness
- Executive Group

ID	Requirement	Implementation / Rationale
HSE-RPT-001	Every metric definition must be visible and versioned.	Prevents ambiguous KPIs.
HSE-RPT-002	Reports must support source-record drill-down and export to PDF/Excel.	Operational verification.
HSE-RPT-003	Historical reporting must use snapshot or	Reproducible trends.

ID	Requirement	Implementation / Rationale
	effective-date logic.	
HSE-RPT-004	Analytics must not automatically label individual employees as unsafe or low-performing.	Responsible use.

37. Platform Configuration

- Module/page/section/widget/action enablement
- Hazard categories
- Risk matrices and severity scales
- Incident and permit types
- Inspection templates
- Induction and toolbox templates
- PPE types
- Emergency scenarios
- Environmental aspects
- Corrective-action types
- Stop-work authority
- Numbering formats
- Review periods
- Competence and eligibility rules
- Confidentiality levels
- Retention rules
- Custom fields
- Notification and approval hooks
- Report templates and KPI definitions

ID	Requirement	Implementation / Rationale
HSE-CONF-001	Configuration changes must be effective-dated and audited.	Historical logic must remain interpretable.
HSE-CONF-002	Disabled features must disappear cleanly from navigation, layouts, search and APIs.	Consistent product behavior.
HSE-CONF-003	Template revisions must not alter completed records.	Reproducibility.
HSE-CONF-004	Critical safety defaults require explicit administrative confirmation before weakening.	Prevent unsafe configuration.

38. UX and Page Architecture

38.1 Design system

- Premium enterprise visual hierarchy
- Status and severity badges with text and icon
- Source-record deep links
- Persistent context header
- Activity timeline
- Responsive data tables and cards
- Command/search actions
- Skeleton loading
- Optimistic UI only for non-critical drafts
- Reduced-motion support

38.2 Critical interaction rules

Interaction	Rule
Issue stop work	Always visible to authorized users in relevant site/work contexts; confirmation includes scope and immediate reason.
Activate permit	Server validation summary must be shown before commit;

Interaction	Rule
	failures identify exact unmet gates.
Close incident/action	Require closure checklist and missing-data explanation.
Critical alert	Cannot rely only on color; must include label, icon and readable text.
Destructive action	Archive/cancel only; no permanent deletion of governed records.
Hover/motion	Clarify hierarchy and status, never delay safety work.

39. Responsive, Mobile and Offline

39.1 Mobile critical actions

- Report hazard/incident/near miss
- Create observation
- Issue stop work
- Scan site/zone QR
- View active permits
- Complete inspection
- Capture evidence
- Record toolbox/induction attendance
- Verify worker eligibility
- Open emergency contacts and plans

39.2 Offline-supported drafts

- Inspection
- Hazard
- Observation
- Initial incident report
- Toolbox attendance
- Induction attendance
- Photos/signatures
- Permit checklist preparation

39.3 Not finalized offline

- Permit activation or closure
- Stop-work release
- Incident closure
- Risk-assessment release
- Emergency resolution

ID	Requirement	Implementation / Rationale
HSE-OFF-001	Offline records must preserve local creation time, device identifier and synchronization time.	Audit of delayed sync.
HSE-OFF-002	Critical offline submissions must be visibly pending until server acknowledgement.	No false completion.
HSE-OFF-003	Conflict resolution must never silently discard field evidence.	Data integrity.
HSE-OFF-004	Cached emergency information must	Reliability.

ID	Requirement	Implementation / Rationale
	have visible last-updated time.	

40. Accessibility

- WCAG 2.2 AA target
- Keyboard operability
- Screen-reader labels and landmarks
- Visible focus
- Text/icon redundancy for status
- Contrast compliance
- Accessible tables
- Reduced motion
- Touch targets suitable for gloves where practical
- Error summaries with field-level links

ID	Requirement	Implementation / Rationale
HSE-A11Y-001	Severity and status must never be conveyed by color alone.	Safety-critical accessibility.
HSE-A11Y-002	Mobile forms must support large touch targets and clear validation.	Field usability.
HSE-A11Y-003	Generated PDFs must use headings, readable tables and logical reading order where supported.	Accessible reporting.

41. Security, Privacy and Tenant Isolation

41.1 Scope model

- Tenant
- Corporate group
- Company
- Project
- Site
- Contractor
- HSE role
- Investigation membership
- Record confidentiality
- Explicit participant

41.2 Sensitive sections

- Injured-person identity
- Medical treatment
- Witness statements
- Legal notes
- Disciplinary links
- Compensation
- Investigation conclusions
- Regulator correspondence

41.3 Capability examples

Capability key	Purpose
hse.view	View standard HSE records in authorized scope.
hse.manage	Create and update operational records.
hse.issue_permit	Issue/activate configured permit types.
hse.stop_work	Create stop-work orders.
hse.release_hold	Release safety holds within authority.
hse.view_sensitive_incident	View restricted personal and investigation sections.
hse.investigate	Create and manage investigations.
hse.close_incident	Close incidents after validation.
hse.configure	Change HSE configuration and templates.
hse.report_export	Export reports and evidence packages.

ID	Requirement	Implementation / Rationale
HSE-SEC-001	All queries must enforce tenant isolation at the database or repository layer.	Prevent cross-tenant leakage.
HSE-SEC-002	Sensitive reads must be audit-ready.	Accountability.
HSE-SEC-003	External contractor users must see only explicitly authorized records and fields.	Third-party privacy.
HSE-SEC-004	Authorization must be enforced by backend regardless of frontend visibility.	Security boundary.
HSE-SEC-005	Evidence download links must be short-lived and permission-checked.	File protection.

42. Functional Data Model

Entity	Purpose
HSESite	Project/site identity, status, leadership, emergency state and configuration.
HSEPlan	Plan lifecycle, applicability, revision and document links.
Hazard / HazardCategory	Reusable hazard definitions and scoped hazard records.
RiskAssessment / RiskAssessmentHazard / RiskControl	Assessment versions, scores, controls and residual risk.
MethodStatementSafetyLink	Applicability and readiness link to Documents record.
PermitType / PermitToWork / PermitWorker / PermitAsset / PermitIsolation / PermitTest	Permit configuration and operational lifecycle.
SafetyInspection / InspectionTemplate / InspectionResponse	Versioned templates and completed inspection data.
SafetyObservation	Unsafe, positive and improvement observations.
Incident / IncidentPerson / IncidentWitness / IncidentAsset / IncidentEvidence	Incident core and linked context.
IncidentInvestigation / RootCause	Investigation governance, findings and cause analysis.
CorrectiveAction / EffectivenessReview	Required actions, Task links, verification and effectiveness.
StopWorkOrder	Hold scope, controls and release history.
Induction / InductionAttendance	Content revision, attendance and eligibility impact.
ToolboxTalk / ToolboxAttendance	Topic, participation and follow-up.
CompetenceRequirement / WorkerEligibility	Operational rules and calculated readiness projection.
PPERequirement / PPEAssignment	Requirements and assignment/compliance references.
EmergencyPlan / EmergencyContact / EmergencyDrill / MusterPoint	Emergency preparedness and response.
EnvironmentalAspect / EnvironmentalMonitoring / EnvironmentalEvent / WasteRecord	Environmental control domain.
ChemicalRegister	Hazardous-substance safety context and SDS link.
ContractorHSEProfile / ContractorHSEEvaluation	Contractor readiness and scorecard.
OccupationalExposure	Operational exposure and restricted status.
HSEDocumentLink / HSEActivity / HSEMetricSnapshot	Cross-domain links, audit timeline and reporting snapshots.

42.1 Common entity fields

- id (UUID)
- tenant_id
- company_id
- project_id/site_id where applicable
- business_code
- status
- version
- created_by/created_at
- updated_by/updated_at
- effective_from/effective_to where applicable
- archived_at/archive_reason
- confidentiality_level
- source_correlation_id

42.2 Data integrity rules

- Immutable issued/final records
- Optimistic version control
- No hard deletion of governed records
- Effective-dated configuration snapshots
- Stable business identifiers
- Tenant-scoped unique constraints
- Foreign-key integrity and controlled archive behavior

43. API Requirements

Endpoint	Purpose
GET /hse/sites	List sites and readiness summaries.
GET /hse/sites/{id}	Open Site HSE Page data.
POST /hse/plans	Create plan.
POST /hse/plans/{id}/issue	Issue current plan revision.
GET /hse/hazards	Search hazard register.
POST /hse/hazards	Create hazard.
POST /hse/risk-assessments	Create assessment.
POST /hse/risk-assessments/{id}/release	Release immutable assessment revision.
POST /hse/risk-assessments/{id}/revise	Create revision.
GET /hse/permits	List/filter permits.
POST /hse/permits	Create permit.
POST /hse/permits/{id}/issue	Issue permit.
POST /hse/permits/{id}/activate	Run gates and activate.
POST /hse/permits/{id}/suspend	Suspend.
POST /hse/permits/{id}/revalidate	Revalidate after suspension/extension.
POST /hse/permits/{id}/close	Close and hand back area.
POST /hse/inspections	Create inspection.
POST /hse/inspections/{id}/responses	Save responses.
POST /hse/inspections/{id}/submit	Submit and apply configured consequences.
POST /hse/observations	Quick observation capture.
POST /hse/incidents	Create initial incident report.
POST /hse/incidents/{id}/triage	Set severity and response.
POST /hse/incidents/{id}/investigations	Create investigation.
POST /hse/incidents/{id}/close	Validate and close incident.
POST /hse/corrective-actions	Create action.
POST /hse/corrective-actions/{id}/convert-to-task	Idempotent Task conversion.
POST /hse/corrective-actions/{id}/verify	Verify/effectiveness result.
POST /hse/stop-work-orders	Issue hold.
POST /hse/stop-work-orders/{id}/release	Validate and release.
POST /hse/inductions	Create induction event.

Endpoint	Purpose
POST /hse/toolbox-talks	Create toolbox talk.
GET /hse/workers/{id}/eligibility	Return operational eligibility and reasons.
POST /hse/emergencies/activate	Activate emergency mode.
POST /hse/emergencies/{id}/resolve	Resolve emergency.
POST /hse/emergency-drills	Record drill.

43.1 API contract rules

- All writes require idempotency key where duplicate submission is possible.
- Version header or payload required for mutable record updates.
- Errors use stable machine codes and user-readable messages.
- List endpoints support pagination, sort, filters, saved-view compatible query model and field projection.
- Critical commands return validation-gate details.
- Bulk actions are asynchronous when operation size exceeds configured threshold.
- All responses include tenant-safe IDs and permission-filtered fields.

44. Backend Services and Domain Events

44.1 Recommended services

- HSE Site Service
- Hazard Register Service
- Risk Assessment Service
- Permit Service
- Safety Gate Service
- Inspection Service
- Observation Service
- Incident Service
- Investigation Service
- Corrective Action Service
- Stop-Work Service
- Training Eligibility Service
- PPE Compliance Service
- Emergency Service
- Environmental Service
- Contractor HSE Service
- Metrics Service
- Document/Task/Asset/Work Progress/HR/Calendar adapters

44.2 Event pattern

- Transactional outbox
- Idempotent consumers
- Correlation and causation IDs
- Actor identity
- Tenant/company/project scope
- Source aggregate/version
- Retry with dead-letter handling
- Observable processing status

44.3 Event catalog

Event	Purpose
HazardReported	New hazard requires triage.

Event	Purpose
RiskAssessmentReleased	Safety controls become applicable.
RiskAssessmentReviewDue	Review/reminder projection.
PermitIssued	Permit available but not active.
PermitActivated	Work may proceed within permit scope.
PermitSuspended	Affected work gate becomes blocked.
PermitExpired	Automatic invalidation.
InspectionFailed	Create configured actions/holds.
CriticalIncidentReported	Emergency/escalation trigger.
IncidentClosed	Final metrics and reports update.
CorrectiveActionOverdue	Escalation-ready event.
StopWorkIssued	Block affected work/assets/permits.
StopWorkReleased	Recalculate readiness.
WorkerEligibilityChanged	Recalculate assignments and permit gates.
AssetComplianceExpired	Recalculate affected work readiness.
EnvironmentalLimitExceeded	Critical event/action/hold.
EmergencyActivated	Activate emergency interface.
HSEMetricSnapshotCreated	Historical reporting.

45. Validation and Edge Cases

Domain	Scenario	Required behavior
Permit	Activation attempted with expired risk assessment.	Reject with gate details; retain Draft/Issued state.
Permit	Two conflicting permits activated concurrently.	Database-backed conflict check and transaction lock; only one succeeds.
Permit	Asset becomes unsafe during active permit.	Suspend/flag affected permit according to configuration and emit events.
Incident	Reporter is offline and syncs after another report exists.	Duplicate detection and merge review; do not discard either record.
Incident	Severity upgraded after executive report generated.	Preserve prior snapshot; future reports use current severity.
Corrective Action	Linked Task completed but evidence is insufficient.	Action remains Under Verification or becomes Ineffective.
Stop Work	Initiator loses access before release.	Hold remains active; authorized release role may continue.
Worker	Employee changes employer during active project.	Historical attendance/employer snapshot retained; current eligibility recalculated.
Risk Assessment	Configuration matrix changes.	Existing assessment retains matrix snapshot.
Documents	Method statement revision archived.	Dependency warning and readiness recalculation.
Emergency	Two users activate same emergency.	Idempotent command returns same active emergency where match rules apply.
Privacy	Unauthorized manager requests medical detail.	Return eligibility only; log denied access where required.
Project	Project archived with open incidents.	Block or require governed closure/migration decision.
Time	Permit spans daylight-saving change.	Store UTC plus source timezone; display correctly.
Offline	Stop-work report created offline while work continues.	Mark pending locally; sync immediately when possible; user sees not server-confirmed warning.

46. Performance, Scalability and Observability

Area	Target / Requirement
Page load	Primary dashboard and list initial response P95 <= 2.5 seconds under agreed reference load, excluding large file previews.
Commands	Critical commands acknowledge server result P95 <= 2 seconds except document processing and large evidence uploads.
Search	Indexed HSE search P95 <= 1.5 seconds for standard tenant scale.
Concurrency	Permit activation and resource/scope conflicts protected by transactional validation.

Area	Target / Requirement
Scale	Support multi-company groups, thousands of users, hundreds of sites and millions of activity/evidence links.
Evidence	Chunked/resumable upload for large files; virus scanning and asynchronous preview generation.
Observability	Structured logs, traces, command latency, event lag, failed consumer counts, sync conflicts and readiness-calculation metrics.
Recovery	Background jobs are idempotent and replayable; dead-letter queues have administrative remediation.
Availability	Critical read workflows designed for high availability; offline mobile provides emergency/readiness resilience.

46.1 Audit requirements

- Create/update/status events
- Before/after values for governed fields
- Actor and delegated actor
- Time and timezone
- IP/device where appropriate
- Correlation ID
- Source module/event
- Sensitive-read audit readiness
- Export and evidence download events

47. Testing Strategy

47.1 Test layers

- Unit tests
- Domain lifecycle tests
- API contract tests
- Integration/event tests
- Permission/privacy tests
- Concurrency tests
- Offline synchronization tests
- Responsive/mobile tests
- Accessibility tests
- Performance/load tests
- Security tests
- Data migration tests
- PDF/report verification

47.2 Critical test matrix

Test area	Scenario	Expected result
Permit activation	Expired operator certificate	Blocked; exact reason returned.
Permit concurrency	Two conflicting permits	One transaction succeeds; one receives conflict.
Asset integration	Unsafe asset on active work	Readiness becomes blocked; configured permit action occurs.
Stop work	Offline field report	Pending local state then immutable server record on sync.

Test area	Scenario	Expected result
Incident evidence	Upload after investigation starts	Evidence appended with integrity metadata.
Worker change	Employer changes mid-project	Historical snapshot remains; current eligibility recalculated.
Risk revision	Revision during active permit	Permit references original; review/revalidation rule triggered.
Privacy	Unauthorized medical access	Denied; no sensitive payload leakage.
Action effectiveness	Task complete but action ineffective	Action reopens/additional action possible.
Emergency	Simultaneous activation	Single active emergency or controlled multiple events by scenario rules.
Tenant isolation	Cross-tenant ID enumeration	No data returned; authorization enforced.
Accessibility	Critical status without color	Text/icon accessible representation exists.

48. Acceptance Criteria

ID	Acceptance criterion
AC-001	HSE Dashboard supports drill-down, filters and critical alerts.
AC-002	Each active site has a complete Site HSE workspace.
AC-003	HSE Plans are revision-controlled and linked to controlled Documents.
AC-004	Hazards are location- and activity-aware with immutable history.
AC-005	Risk assessments support initial/residual risk, control hierarchy and revisions.
AC-006	Permit activation executes all configured safety gates server-side.
AC-007	Permit conflicts, expiry, suspension and closure are fully enforced.
AC-008	Inspection templates are versioned and critical failures trigger configured consequences.
AC-009	Mobile observations support rapid capture and immediate-danger escalation.
AC-010	Incidents support quick initial reporting, triage, investigation, action and closure.
AC-011	Sensitive incident and health data is field-level permission ready.
AC-012	Investigation evidence preserves integrity metadata and immutable originals.
AC-013	Corrective actions integrate with Tasks without surrendering HSE verification authority.
AC-014	Stop-work orders visibly block affected records and require governed release.
AC-015	Induction and competence data produces operational worker eligibility.

ID	Acceptance criterion
AC-016	PPE compliance integrates with Inventory and Assets without duplication.
AC-017	Emergency plans, contacts, drills and active emergency mode are operational.
AC-018	Environmental aspects, monitoring, waste and events are supported.
AC-019	Chemical records link controlled SDS documents and operational restrictions.
AC-020	Contractor HSE profiles support prequalification, performance and restrictions.
AC-021	Work Progress receives reliable safety-readiness status.
AC-022	Assets integration responds to compliance expiry and unsafe outcomes.
AC-023	Calendar projections link back to HSE source records.
AC-024	Notification and approval integration hooks are defined but not duplicated.
AC-025	Global search respects all permission and confidentiality rules.
AC-026	Reports expose metric definitions and drill down to source records.
AC-027	Every feature supports Platform Configuration.
AC-028	Mobile and offline drafts synchronize without silent evidence loss.
AC-029	Critical safety warnings satisfy accessibility requirements.
AC-030	All backend endpoints enforce tenant isolation and authorization.
AC-031	Governed records cannot be hard deleted.
AC-032	Audit history captures status, actor, time, reason and correlation data.
AC-033	APIs support idempotency and optimistic concurrency.
AC-034	Event consumers are idempotent and observable.
AC-035	Performance targets are validated under reference load.
AC-036	All critical test scenarios pass before production release.

49. Delivery Phases

49.1 Phase 1 - Operational Foundation

- HSE Dashboard and Site HSE Page
- Hazards and Risk Assessments
- Method Statement links
- Permits
- Inspections and Observations
- Incidents, Investigations and Actions
- Stop Work
- Inductions and Toolbox Talks
- Emergency contacts
- Documents, Tasks and Assets integration
- Responsive mobile workflows
- Audit and core reports

49.2 Phase 2 - Advanced Control

- PPE and worker eligibility
- Contractor HSE
- Environmental management
- Chemical register
- Emergency drills and advanced emergency mode
- Advanced analytics
- Offline operations
- Work Progress safety gates
- Calendar integration

49.3 Future Roadmap

- IoT environmental sensors
- Wearables and geofencing
- Automatic muster
- Telematics
- AI-assisted hazard detection
- Image-based PPE detection
- Predictive analytics
- External regulatory integrations
- Video training integrations

Initial-release rule

External integrations must remain optional. The core HSE platform must be fully usable as a self-contained Construction OS module.

50. Page and Flow Registry

ID	Page	Route
HSE-001	HSE Dashboard	/hse
HSE-002	Sites	/hse/sites
HSE-003	Site HSE Page	/hse/sites/{siteId}
HSE-004	HSE Plans	/hse/plans
HSE-005	Hazards	/hse/hazards
HSE-006	Risk Assessments	/hse/risk-assessments

ID	Page	Route
HSE-007	Permits	/hse/permits
HSE-008	Permit Page	/hse/permits/{permitId}
HSE-009	Inspections	/hse/inspections
HSE-010	Observations	/hse/observations
HSE-011	Incidents	/hse/incidents
HSE-012	Incident Page	/hse/incidents/{incidentId}
HSE-013	Investigations	/hse/investigations
HSE-014	Corrective Actions	/hse/actions
HSE-015	Stop Work	/hse/stop-work
HSE-016	Inductions	/hse/inductions
HSE-017	Toolbox Talks	/hse/toolbox-talks
HSE-018	Training and Competence	/hse/competence
HSE-019	PPE	/hse/ppe
HSE-020	Emergency	/hse/emergency
HSE-021	Environment	/hse/environment
HSE-022	Chemicals	/hse/chemicals
HSE-023	Contractor HSE	/hse/contractors
HSE-024	Reports	/hse/reports
HSE-025	Settings	/hse/settings

50.1 Core flows

- Report Hazard
- Create/Release Risk Assessment
- Prepare/Issue/Activate/Suspend/Close Permit
- Complete Inspection
- Report Observation
- Report/Triage/Investigate/Close Incident
- Create/Convert/Verify Corrective Action
- Issue/Release Stop Work
- Record Induction
- Record Toolbox Talk
- Activate/Resolve Emergency
- Report/Close Environmental Event
- Evaluate Contractor HSE

51. Risks and Assumptions

Risk / Assumption	Mitigation
Jurisdiction-specific safety rules vary.	Use configurable types, matrices, templates and approval hooks; avoid embedding legal advice.
Poor field connectivity.	Offline drafts, cached emergency data, resumable uploads and clear pending state.
Overly complex forms reduce reporting.	Progressive disclosure, templates and minimum safe initial report.

Risk / Assumption	Mitigation
Sensitive data leakage.	Field-level permissions, backend enforcement, audit and tenant isolation.
Duplicated ownership across modules.	Strict source-of-truth rules and event-based projections.
Configuration weakens controls.	Audit, effective dating and confirmation for critical defaults.
Safety metrics used as simplistic employee scoring.	Source drill-down, contextual reporting and explicit non-goal.
False confidence from projected headcount/readiness.	Freshness labels, calculation details and authoritative-source indicators.

52. Final Implementation Checklist

- ☐ 01. Domain ownership reviewed with Projects, Work Progress, QA/QC, Assets, HR, Tasks, Documents, Contracts, Inventory and Finance.
- ☐ 02. All lifecycle transitions implemented server-side.
- ☐ 03. Immutable/revision rules enforced.
- ☐ 04. Authorization and tenant isolation covered by automated tests.
- ☐ 05. Critical validation gates implemented atomically.
- ☐ 06. Idempotency and concurrency controls verified.
- ☐ 07. Platform Configuration applied to all pages and actions.
- ☐ 08. Mobile critical workflows tested on real devices.
- ☐ 09. Offline conflict behavior verified.
- ☐ 10. Accessibility audit completed.
- ☐ 11. Event catalog and consumers documented.
- ☐ 12. Audit logging and sensitive-read strategy implemented.
- ☐ 13. Report definitions and snapshot logic approved.
- ☐ 14. Performance/load tests passed.
- ☐ 15. Data retention and archive behavior approved.
- ☐ 16. Operational runbooks and monitoring dashboards prepared.

Final product position

The HSE Module is the live operational control system that determines whether people, equipment and construction activities are safe and environmentally compliant before, during and after work.

END OF PRD_HSE_MODULE v1.0